MinJae Kim

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EDUCATION

Sep. 2025 - **Stanford University**
PhD Student, Department of Materials Science and Engineering
(Advisor: Prof. Guosong Hong and Prof. Mark Brongersma)

Feb. 2019 - Feb. 2025 **Korea Advanced Institute of Science and Technology (KAIST)**
Bachelor of Science in Materials Science and Engineering

- Total GPA of 4.22/4.3 (99.2/100), *Summa Cum Laude*, Honor student
- Valedictorian and Representative Graduate at KAIST Commencement 2025 (Speech video)
- Global Leadership Award (Top 18 out of ca. 11,000 students)
- College of Engineering Leadership Award (Top 10 out of ca. 3,000 students)
- Dean's List (Top 3% in College of Engineering)
- Departmental Honor Scholarship (Top 4 out of ca. 120 students)
- Fulfilled obligatory military service in Republic of Korea Army (Mar. 2021 - Sep. 2022)

PUBLICATIONS

2. M.J. Kim†, J. Kim†, S. Yoo*, **Near-planar light outcoupling structures with finite lateral dimensions for ultra-efficient and optical crosstalk-free OLED displays**, *Nat. Commun.*, 2025, DOI: 10.1038/s41467-025-66538-6

1. M.J. Kim†, D. Choi†, C. Kang, S. Yoo*, **Ultralow-power carbon dioxide sensor for real-time breath monitoring**, *Device*, 2025, DOI: 10.1016/j.device.2024.100681

HONORS AND AWARDS

Scholarships

2025 - 2030	**Kwanjeong Doctoral Study Abroad Fellowship**, Kwanjeong Foundation
2023 - 2024	**Woonhae Scholarship**, Woonhae Foundation
2023	**Young-Han Kim Global Leader Scholarship**, KAIST
2022 - 2026	**Dream Supporter Scholarship**, Global Hansang Dream Foundation
2021 - 2025	**KAIST Presidential Fellowship**, KAIST
2019 - 2025	**National Presidential Science Scholarship**, President of South Korea

Honors and Awards

2026	**First Place, Falling Walls Lab San Francisco Bay Area**, German Consulate General San Francisco
2026	**National Delegate to Global Young Scientists Summit**, National Research Foundation of Singapore
2025	**Valedictorian at KAIST Commencement 2025**, KAIST
2024	**National Delegate to 73rd Lindau Nobel Laureate Meeting**, Korean Academy of Science and Technology
2024	**NUS Young Fellow**, National University of Singapore
2023	**Young Future Energy Leader**, Khalifa University
2023	**Representative of KAIST, Young Engineers Honor Society**, National Academy of Engineering of Korea
2022	**Bronze Award**, Asian Students' Venture Forum
2021	**Talent Award of Korea**, Ministry of Education
2020	**Nobel Ceremony Guest and National Delegate**, Stockholm International Youth Science Seminar (SIYSS)
2020	**Cadet, Research Officer for National Defense**, Ministry of Defense, Ministry of Science and ICT

PATENTS

5. M.J. Kim, S. Yoo*, **Optical film comprising plurality of light-absorbing elements, and light-emitting diode comprising optical film** (KR-10-2025-0197889)

4. M.J. Kim, S. Yoo*, **Light emitting device comprising optical structure minimizing angular color shift and method for forming the structure thereof** (KR10-2025-0117210)

3. M.J. Kim, S. Yoo*, **Light emitting device for efficient light extraction and method for determining its structure** (KR10-2024-0201956)

2. M.J. Kim, H. Cho*, **Perovskite material including multi-functional asymmetric organic spacer and light-emitting device including the same** (KR10-2023-0118990)

1. M.J. Kim, S. Shin, H. Cho*, **Semiconductor material with passivation molecule giving π-conjugation electron density and method for manufacturing the same** (KR10-2023-0167999)

PRESENTATIONS

5. M.J. Kim, J. Kim, S. Yoo*, **Near-planar light outcoupling structure for ultra-efficient organic light-emitting diodes**, *UC Berkeley-Stanford Academic Conference* (Oral), 2025
   Best Presentation Award

4. M.J. Kim, J. Kim, S. Yoo*, **Ultralow-power and highly stable carbon dioxide sensor for real-time breath monitoring**, *KAIST URP Workshop* (Oral), 2024
   Grand Prix

3. M.J. Kim, J. Kim, S. Yoo*, **Near-planar light outcoupling structure for ultra-efficient organic light-emitting diodes**, *Optics and Photonics Congress* (Oral), 2024
   Best Paper Award

2. M.J. Kim, S. Shin, H. Cho*, **Highly luminescent and stable quasi-2D perovskites based on multi-functional asymmetric spacer**, *KAIST URP Workshop* (Oral), 2023
   Grand Prix

1. M.J. Kim, S. Shin, H. Cho*, **Highly luminescent and stable quasi-2D perovskites based on multi-functional asymmetric spacer**, *Spring Meeting of Korea Institute of Metals and Materials* (Poster), 2023
   Best Poster Presentation Award

SKILLS

Language	English (fluent, TOEFL iBT: 106), Korean (native)
	LaTeX(advanced), MATLAB (advanced), Python (moderate), HTML (moderate)
Simulation	LightTools (advanced), ORCA Density Functional Theory (intermediate), COMSOL Multiphysics (intermediate), RCWA (intermediate) Lumerical (novice)
Technical	Metasurface and plasmonics design and fabrication, Optical and photonic design of optoelectronics, PeLED/OLED fabrication and characterization, Organic synthesis and analysis

REFERENCE

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